

# CBSE Practice Worksheet

Class 8 | Mathematics | Difficulty: Hard | Set E

**Instructions:** Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

## Section A: Multiple Choice Questions (1 mark each)

1. Factorize:  $x^2 - 5x - 14$   
(A)  $(x-7)(x+2)$   
(B)  $(x+7)(x-2)$   
(C)  $(x-7)(x-2)$   
(D)  $(x+7)(x+2)$
2. Solve:  $2^{(x+1)} = 32$   
(A)  $x = 3$   
(B)  $x = 4$   
(C)  $x = 5$   
(D)  $x = 6$
3. If  $a:b = 3:4$  and  $b:c = 5:6$ , find  $a:c$   
(A)  $3:6$   
(B)  $5:8$   
(C)  $15:24$   
(D)  $1:2$
4. Simplify:  $(x^3 - 8)/(x - 2)$   
(A)  $x^2 + 2x + 4$   
(B)  $x^2 - 2x + 4$   
(C)  $x^2 + 4$   
(D)  $x^2 - 4$
5. Solve:  $3(x-2)/4 = (2x+1)/3$   
(A)  $x = 14$   
(B)  $x = 22$   
(C)  $x = 26$   
(D)  $x = 30$
6. Solve:  $|2x - 5| = 9$   
(A)  $x = 7$  or  $-2$   
(B)  $x = 7$  or  $2$   
(C)  $x = -7$  or  $2$   
(D)  $x = -7$  or  $-2$
7. If  $x + 1/x = 5$ , find  $x^2 + 1/x^2$   
(A) 21  
(B) 23  
(C) 25  
(D) 27
8. What is the cube root of -125?  
(A) 5  
(B) -5  
(C)  $\pm 5$

(D) undefined

9. Simplify:  $(a^2 - b^2)/(a + b)$

(A)  $a - b$

(B)  $a + b$

(C)  $a^2 - b$

(D)  $a - b^2$

10. If  $\log_{\blacksquare} 8 = x$ , find  $x$

(A) 2

(B) 3

(C) 4

(D) 8

### Section B: Fill in the Blanks (1 mark each)

11. If  $x + y = 7$  and  $xy = 12$ , then  $x^2 + y^2 = \underline{\hspace{2cm}}$

12.  $\text{HCF} \times \text{LCM}$  of 12 and 18 =  $\underline{\hspace{2cm}}$

13. Rationalize:  $1/(\sqrt{3} + 1) = (\sqrt{3} - 1)/\underline{\hspace{2cm}}$

14. If  $a + b + c = 0$ , then  $a^3 + b^3 + c^3 = \underline{\hspace{2cm}}$

15.  $(x + 1/x)^2 - (x - 1/x)^2 = \underline{\hspace{2cm}}$

16. If  $x^2 - 6x + 8 = 0$ , then  $x = \underline{\hspace{2cm}}$  or  $\underline{\hspace{2cm}}$

17.  $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - \underline{\hspace{2cm}}$

18. If  $5^{\blacksquare} = 125$ , then  $x = \underline{\hspace{2cm}}$

### Section C: Short Answer Questions (2 marks each)

19. Factorize:  $8x^3 - 27$

20. Solve:  $|x - 3| + |x + 2| = 7$

21. If  $x = 2 + \sqrt{3}$ , find  $x + 1/x$

22. Solve:  $\log_{\blacksquare}(x - 1) + \log_{\blacksquare}(x + 1) = 3$

23. Simplify:  $(x^{\blacksquare} - 16)/(x^2 + 4)$

24. Simplify:  $(2 + \sqrt{3})/(2 - \sqrt{3})$

25. Solve:  $x^2 - 7x + 12 = 0$

--- End of Worksheet ---

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